

TrES-3b

R-0277

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180521 · created 2026-07-17T14:30:34+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458259.9081 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1963 +/- 0.0091
Transit depth (Rp/Rs)^2	3.85 +/- 0.36 [%]
Orbital Inclination (inc)	82.54 +/- 0.4
Airmass coefficient 1 (a1)	1.237 +/- 0.016
Airmass coefficient 2 (a2)	-0.2 +/- 0.065
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	#3 - [364, 22]
Optimal Aperture	3.59
Optimal Annulus	5.93
Transit Duration (day)	0.0643 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1963 ± 0.0091 vs 0.16309 (deviation 3.24σ)
Duration fit vs published	0.0643 d vs 0.0567 d (deviation 2.26σ)
Mid-time O–C	2.0 min
Depth SNR	19.1
Residual scatter	1.26 %

Light curve

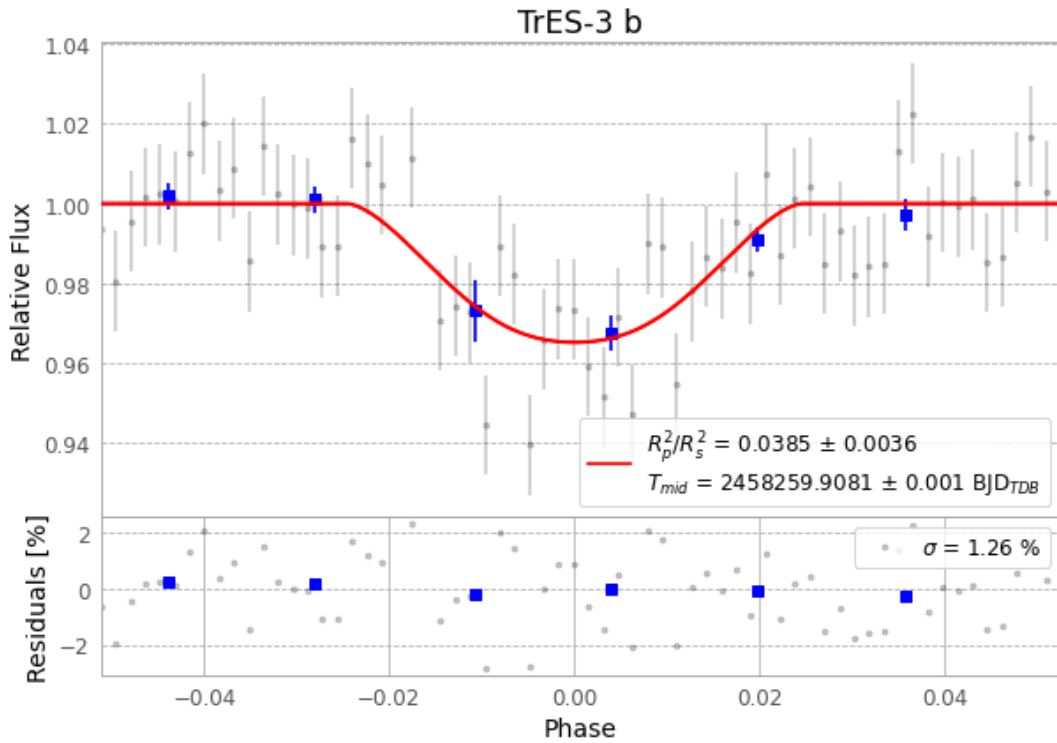


Figure 1 — FinalLightCurve_TrES-3 b_2018-05-21.png (SHA-256 2baa6a19a5aaa66d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2077c3f9aace8885...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8d12984

Data provenance

66 FITS frames (43 MB), TRES-3180521080612.FITS → TRES-3180521112112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180521.log (50fc294087c8...), FinalLightCurve_TrES-3 b_2018-05-21.png (2baa6a19a5aa...), FinalLightCurve_TrES-3 b_2018-05-21.csv (dbad1d83e026...)

Compute resources

Wall time 610.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 14:30Z.